

Gurpreet Singh

Fairfax, VA | +1 571 513 1491 | gurpreetkukreja13@gmail.com | [LinkedIn](#) | [Github](#) | [Portfolio](#)

Summary

Software and AI Engineer specializing in full-stack architecture and applied machine learning. Skilled in building enterprise Agentic solutions, deploying predictive models, and developing clean, scalable applications from end to end. Experienced in guiding Agile development teams, managing IT operations, and collaborating across disciplines to deliver practical software solutions

EDUCATION

George Mason University

Aug 2023 - May 2027

Bachelor of Science, Computer Science

Fairfax, VA

- **Coursework:** Operating Systems, Data Structures and Algorithms (C++), Low-Level Programming, Object Oriented Programming, Computer Architecture and Systems Programming, Linear Algebra, Intro to Data Science, Probability and Statistics, Formal Methods and Models, Discrete Mathematics, Multivariable Calculus, Black Holes

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, C++, JavaScript, C, SQL, TypeScript, HTML, CSS, R
- **Frameworks and Tools:** Node.JS, SpringBoot, Flask, NoSQL, Postgres SQL, AWS, AWS Agentcore, AWS Bedrock, React.JS, Angular, Junit, Docker, AI Studio, Azure AD
- **Technologies:** Git, Figma, Linux, Gen AI, Postman, RESTful APIs, CI/CD, Machine Learning, Agile, GNN, Computer Vision, Agentic AI

EXPERIENCE

Presidio Networked Solutions LLC

Jun 2026 - Present

Agentic AI Engineer - Digital Field CTO Intern

Herndon, VA

- Supported the CTO in the development of an enterprise-wide AI control plane, assisting in the creation of foundational governance frameworks to securely scale generative AI solutions.
- Architected custom enterprise AI agents using AWS Bedrock, Agent Core, and AI Studio, formulating comprehensive AI guardrails to ensure secure framework compliance for deployments in healthcare and sports media.
- Developed an intelligent Microsoft Teams agent that automates meeting transcription, context-aware topic extraction, and automated video clipping, successfully filtering sensitive information and reducing manual video editing time by 50%.
- Engineered a multimodal sports analytics agent that aggregates player health metrics, environmental conditions, travel logs, and training data to generate predictive injury risk profiles, optimizing team readiness for high-intensity match schedules.

iCode

Jul 2024 - Present

Software Engineering Technical Lead

Burke, VA

- Developed a new AI and NLP curriculum adopted franchise-wide, alongside coursework in full-stack development, cybersecurity, and Networking using Python, Java, Javascript, HTML/CSS and C++ .
- Engineered and deployed custom software applications for daily classroom use, while mentoring student project teams in Agile methodologies and Git.
- Manage IT operations for Two locations (200+ laptops and systems), including server administration, networking, and technical support to ensure seamless classroom and lab performance.
- Facilitated robotics development integrating hardware, automation, and sensor-driven systems, demonstrating ability to build tools that interact with both hardware and software.

OceanicAI

Dec 2025 - Jun 2026

GNN/CV Fellow(William and Mary)

- Engineered a high-performance Spatio-Temporal Graph Neural Network (GNN) pipeline to analyze a dataset of 237,676 HABCAM underwater images, optimizing marine biomass estimation across the Mid-Atlantic region.
- Architected a GCN-GRU hybrid model that integrated historical temporal context to achieve a 57% reduction in Mean Absolute Error
- Collaborated with federal and academic partners (VIMS, NOAA, NSF) to prototype deep learning defenses against adversarial data poisoning in oceanic computer vision systems.

George Mason University

Aug 2024 - Jun 2026

Research/Learning Assistant

Fairfax, VA

- Applied Python, data visualization, and machine learning to enhance curriculum and improve engagement in data-driven problem solving.
- Debugged and maintained remote observatory systems and simulation software, improving system reliability and performance.
- Resolved hardware/software integration issues across labs and research tools, ensuring smooth operation of astrophysics and engineering projects.
- Collaborated with faculty and students on cross-disciplinary technical challenges, demonstrating independence, problem-solving, and team-oriented communication.

PROJECTS

Inventory & Supply Optimization System

Dec 2025

- Built full-stack system with Spring Boot and React for real-time inventory, supplier management, and restocking.
- Integrated Python AI/ML engine for demand forecasting and supplier optimization, improving timing and cost.
- Designed responsive React frontend with dynamic forms, tables, and interactive charts via Axios.
- Implemented PostgreSQL for data, MongoDB for logs/predictions, Docker for deployment, and Postman for testing.

Autonomous Traffic Flow Optimization Platform

Sep 2025

- Built and presented at the NDIA Department of Defense Hackathon under the Geo-Spatial Analytics & Autonomous Systems.
- Developed a machine learning–optimized traffic signal system using TensorFlow and Python to minimize/eliminate vehicle wait times.
- Trained on FCPS traffic signal timing data, achieving a 65% reduction in wait times (100% in low-traffic autonomous conditions).
- Integrated with autonomous vehicle communication protocols to dynamically adjust speeds and intersection flow.

KodaCare

Mar 2026

- A multimodal health companion designed to automate chronic illness tracking, securing 2nd Place in the Healthcare Track by capturing real-time wellness data to bridge blind spots between physician visits.
- Integrated Gemini 2.0 Pro to perform native multimodal analysis, converting unstructured audio notes and symptom imagery
- Implemented a Singleton Database Manager within a Flask backend to optimize MongoDB transactions and ensure the secure storage of sensitive longitudinal health records.
- Developed an intelligent logic system that automatically organizes disjointed updates into cohesive medical histories for improved diagnostic accuracy.

Data Structures Visualizer

May 2025

- Built cross-platform desktop app animating 8+ data structures and algorithms with step-by-step visual explanations.
- Developed modular visualizers for arrays, linked lists, stacks, queues, and trees to explore core operations interactively.
- Designed modern, scalable UI with tutorials and dynamic controls to enhance algorithm learning.
- Enabled user-driven operations (insert, delete, swap, traverse) with animated feedback to improve comprehension.